

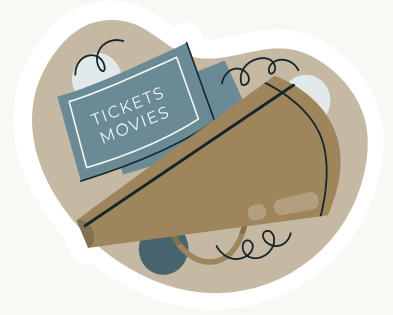


# Movie Box Office Revenue Prediction

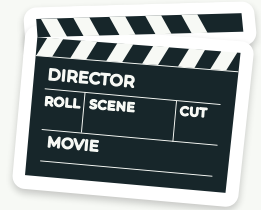
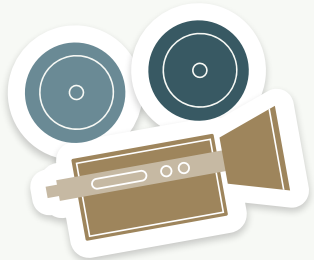
By: Terryl-Errol Foryoung, Nicolas Moscato-Barbeau,  
Gulnoor Kaur, Sofia Valiante, Daniel Pace

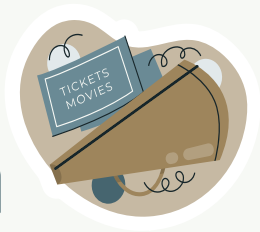


# Topic and Defined ML Tasks



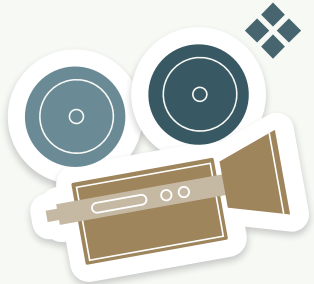
- ❖ The project's goal is to estimate a movie's revenue based on **metadata** through **regression analysis**.
- ❖ Metadata attributes include: budget, genres, release dates, directors, and cast members.
- ❖ The main output is a single value prediction: **box office revenue**.



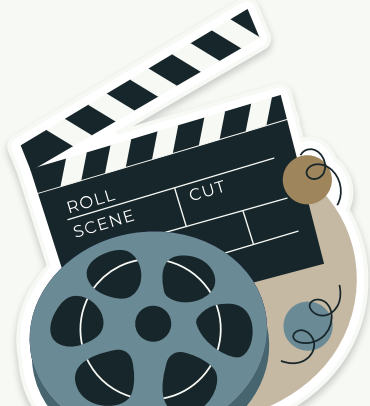
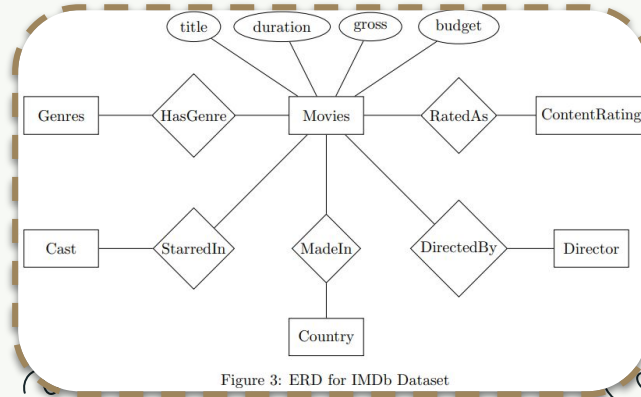
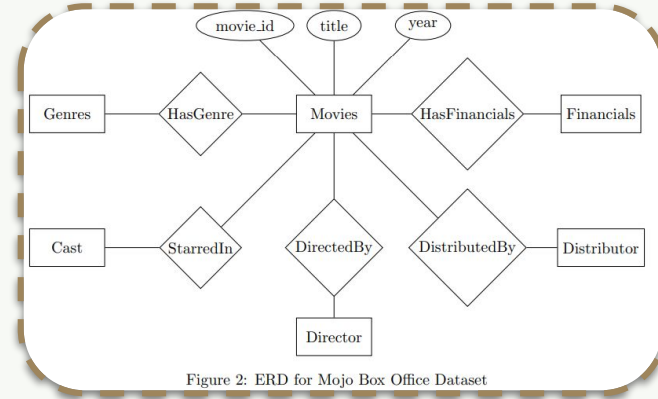
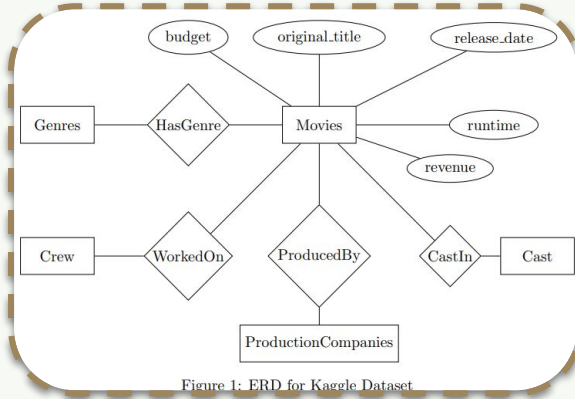


# Datasets - Collections and System

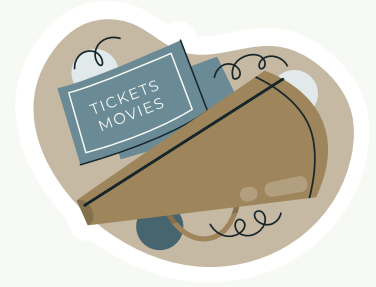
- ❖ Three datasets sourced from **Kaggle**, **IMDb**, and **Box Office Mojo**.
- ❖ Combined total of over 11,000 rows and more than 20 columns of movie metadata.
- ❖ **Significant preprocessing** was done to align the datasets to ensure compatibility for analysis.
- ❖ Storage and processing was handled using **Python**, **Excel**, and **Google Colab** for accessibility and scalability.



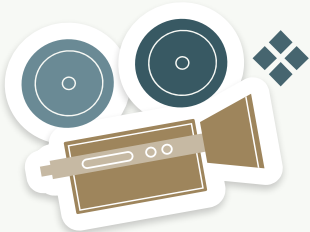
# Entity Relationship Diagrams



# Data Integration

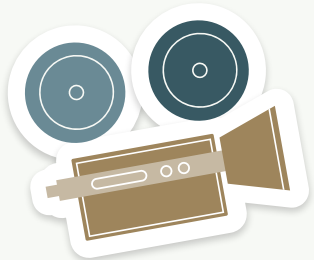
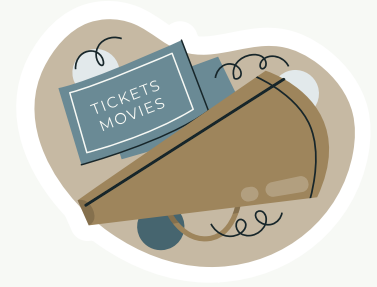


- ❖ **Addressed inconsistencies** in schema formats across datasets.
- ❖ **Standardized formats** for attributes like genres, release dates, and runtimes.
- ❖ **Synced schema** with column mapping between datasets to ensure consistency.
- ❖ **Removed duplicates** by comparing titles and release years, using Python for automation.
- ❖ **Established functional dependencies**, identifying (Title, Year) as a candidate key for integration.

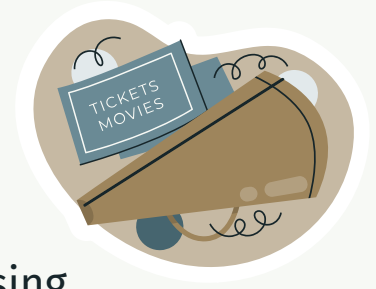


# Column Mapping

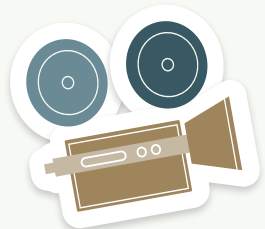
| Mojo         | IMDb           | Kaggle               |
|--------------|----------------|----------------------|
| title        | movie_title    | title                |
| year         | title_year     |                      |
| mpaa         | content_rating |                      |
| release_date |                | release_date         |
| run_time     | duration       | runtime              |
| distributor  |                | production_companies |
| director     | director_name  | crew                 |
| main_actor_1 | actor_1_name   | cast (JSON List)     |
| main_actor_2 | actor_2_name   | cast (JSON List)     |
| main_actor_3 | actor_3_name   | cast (JSON List)     |
| main_actor_4 |                | cast (JSON List)     |
| budget       | budget         | budget               |
| worldwide    | gross          | revenue              |
| genre_1      | genres (List)  | genres (JSON List)   |
| genre_2      | genres (List)  | genres (JSON List)   |
| genre_3      | genres (List)  | genres (JSON List)   |
| genre_4      | genres (List)  | genres (JSON List)   |



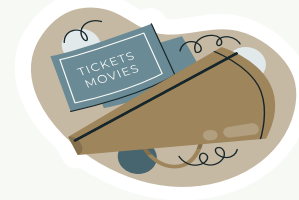
# Data Cleaning



- ❖ **Null values handled** by removing records with critical missing data.
- ❖ Outliers identified using methods like **Z-score normalization** and manual observation.
- ❖ Attributes like budget and revenue **adjusted for inflation** to ensure consistency.
- ❖ Inconsistent or missing runtime and revenue values **removed** as **noisy data**.
- ❖ Standardized formats for categorical attributes like genres through **label encoding**.



# Data Cleaning



**Examples:** Attributes in which the entire record was removed due to a null value that couldn't be supplemented from another dataset were directors, revenue, missing cast, and distributor.

1. The director is missing

|      |            |                    |
|------|------------|--------------------|
| 1982 | 1982-12-07 | Paramount Pictures |
|------|------------|--------------------|

Figure 7: Missing Director

2. The attribute runtime also had values equal to zero which was not realistic for the length of a movie and was identified as noisy data, possibly due to a measuring error. In the example below, the runtime is zero.

|         |          |        |   |       |
|---------|----------|--------|---|-------|
| 6843500 | 10703234 | Comedy | 0 | FALSE |
|---------|----------|--------|---|-------|

Figure 12: Runtime Outlier Example

3. There were cases where revenue was zero. Given that revenue is the most important attribute for our dataset as we wish to predict it, we determined that only precise and qualified revenues would be sufficient, therefore any revenue of value zero was removed.

|      |                  |            |                 |                  |         |   |           |        |        |         |
|------|------------------|------------|-----------------|------------------|---------|---|-----------|--------|--------|---------|
| 2008 | Publico Porester | Alan Daton | Santiago Segura | Veronica Heister | 7800000 | 0 | Adventure | Comedy | Family | Fantasy |
|------|------------------|------------|-----------------|------------------|---------|---|-----------|--------|--------|---------|

Figure 13: Revenue Outlier Example

3. The entire cast is missing

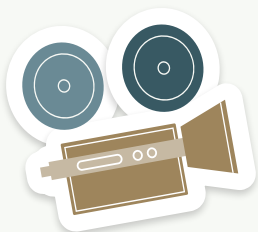
|      |            |                   |
|------|------------|-------------------|
| 2004 | 2004-11-03 | Columbia Pictures |
|------|------------|-------------------|

Figure 9: Missing Cast

4. The distributor is missing

|      |           |              |                |                 |              |
|------|-----------|--------------|----------------|-----------------|--------------|
| 2015 | Not Rated | Tara Subkoff | Timothy Hutton | Balthazar Getty | Lydia Hearst |
|------|-----------|--------------|----------------|-----------------|--------------|

Figure 10: Missing Distributor





# Data Transformation



## Data Normalization

- ❖ All monetary values were adjusted for inflation using the Python package CPI.
- ❖ Standardized date format using regex pattern matching to map to DD-MMM.

## Data Scaling Functions

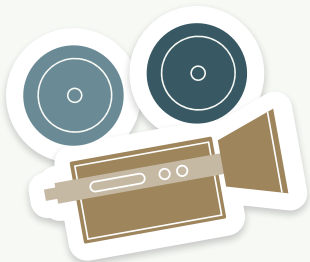
- ❖ Z-score normalization on all continuous attributes 'year', 'budget', 'revenue', 'runtime'. 'Budget' and 'revenue' already adjusted for inflation.
- ❖ Absolute value taken on all results.
- ❖ Z-score outlier detection to filter the set. Any z-score greater than 3 was considered an outlier and removed.

## Data Encoding

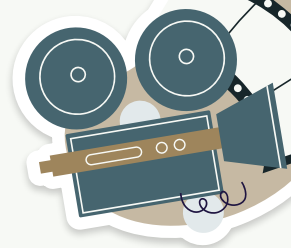
- ❖ Label encoded nominal attributes 'genre\_#', 'mpaa' and 'release date'.
- ❖ Ordinal encoding performed for attributes 'distributor', 'director', 'main\_actor\_#'.

### Ordinal Encoding Method

1. Converted value\_counts of each feature into a dictionary.
2. Mapped dictionary values back to their respective columns.



# Data Transformation



| budget     | revenue     | budget       | revenue      |
|------------|-------------|--------------|--------------|
| 1500000.0  | NaN         | 1.693513e+07 | 8.117258e+07 |
| 0.0        | 1625847.0   | 1.321479e+08 | 3.395374e+08 |
| 7500000.0  | 60722734.0  | 5.645044e+07 | 7.836604e+07 |
| 15000000.0 | 108286421.0 | 9.077398e+07 | 1.388919e+07 |
| 15000000.0 | 71897215.0  | 2.980787e+07 | 1.293259e+08 |

Figure 17: Inflation Adjustment

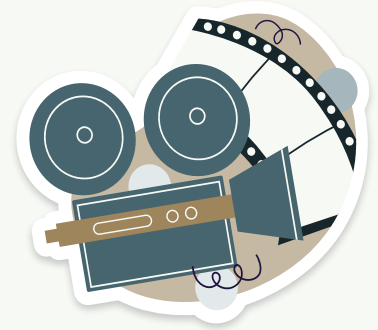
```
distributor
Warner Bros.      378
Universal Pictures 326
Twentieth Century Fox 286
Sony Pictures Releasing 260
Paramount Pictures 241
...
Slugger Pictures  1
Indican Pictures  1
Cowboy Pictures   1
Variance Films    1
Clarius Entertainment 1
Name: count, Length: 162, dtype: int64
```

|   |                     |        |   |     |                    |
|---|---------------------|--------|---|-----|--------------------|
| 7 | Paranormal Activity | 2007.0 | 4 | 299 | Paramount Pictures |
| 7 | Paranormal Activity | 2007.0 | 4 | 299 | 241                |

*Ordinal Encoding*



# Data Transformation



|        |     |     |     |     |     |     |     |     |          |           |
|--------|-----|-----|-----|-----|-----|-----|-----|-----|----------|-----------|
| 2004.0 | 2   | 49  | 3   | 2   | 1   | 9   | 1   | 1.0 | 1.045252 | 0.664990  |
| 1998.0 | 4   | 12  | 1   | 10  | 1   | 1   | 1   | 1.0 | 1.045112 | 0.666069  |
| 2004.0 | 3   | 94  | 6   | 1   | 1   | 1   | 1   | 1.0 | 1.045111 | 0.663512  |
| 2007.0 | 4   | 299 | 241 | 1   | 2   | 2   | 1   | 1.0 | 1.044951 | 0.254377  |
| 1990.0 | 4   | 53  | 1   | 7   | 1   | 1   | 1   | 1.0 | 1.044484 | 0.657086  |
| ...    | ... | ... | ... | ... | ... | ... | ... | ... | ...      | ...       |
| 2019.0 | 3   | 206 | 223 | 6   | 1   | 1   | 1   | 1.0 | 3.810094 | 3.481955  |
| 2018.0 | 3   | 272 | 223 | 13  | 1   | 6   | 2   | 1.0 | 3.898073 | 0.878599  |
| 2007.0 | 3   | 260 | 223 | 10  | 25  | 2   | 4   | 2.0 | 5.485749 | 3.909801  |
| 2017.0 | 3   | 146 | 223 | 3   | 2   | 2   | 1   | 1.0 | 4.792238 | 4.701094  |
| 2019.0 | 3   | 276 | 223 | 4   | 12  | 6   | 8   | 2.0 | 5.240222 | 10.138672 |

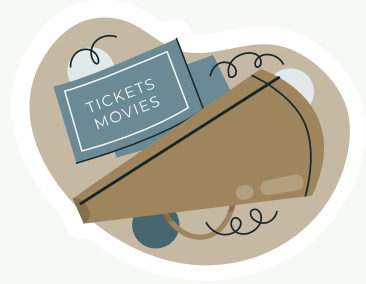
Figure 14: Z-scores for attributes

|        |     |     |     |     |     |     |     |     |          |          |
|--------|-----|-----|-----|-----|-----|-----|-----|-----|----------|----------|
| 2004.0 | 2   | 49  | 3   | 2   | 1   | 9   | 1   | 1.0 | 1.045252 | 0.664990 |
| 1998.0 | 4   | 12  | 1   | 10  | 1   | 1   | 1   | 1.0 | 1.045112 | 0.666069 |
| 2004.0 | 3   | 94  | 6   | 1   | 1   | 1   | 1   | 1.0 | 1.045111 | 0.663512 |
| 2007.0 | 4   | 299 | 241 | 1   | 2   | 2   | 1   | 1.0 | 1.044951 | 0.254377 |
| 1990.0 | 4   | 53  | 1   | 7   | 1   | 1   | 1   | 1.0 | 1.044484 | 0.657086 |
| ...    | ... | ... | ... | ... | ... | ... | ... | ... | ...      | ...      |
| 2014.0 | 3   | 248 | 286 | 9   | 8   | 4   | 4   | 1.0 | 2.768130 | 2.445126 |
| 2017.0 | 3   | 288 | 223 | 4   | 6   | 5   | 2   | 2.0 | 2.637697 | 2.812837 |
| 2018.0 | 3   | 142 | 378 | 5   | 3   | 3   | 4   | 1.0 | 2.549886 | 1.908499 |
| 2019.0 | 3   | 54  | 286 | 1   | 7   | 3   | 3   | 3.0 | 2.485902 | 0.308565 |
| 2017.0 | 3   | 246 | 241 | 13  | 22  | 9   | 4   | 1.0 | 2.950750 | 1.772283 |

Figure 15: Outliers Removed



# Models After Data Cleaning



## Stacked Ensemble

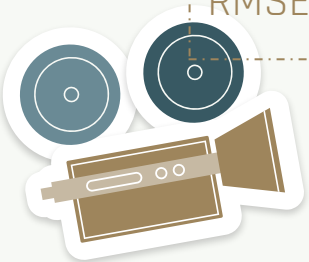
|       |             |
|-------|-------------|
| MSE   | 3.16218e+16 |
| MAE   | 1.0009e+08  |
| RMSLE | nan         |
| RMSE  | 1.77825e+08 |

## Gradient Boosting Machine

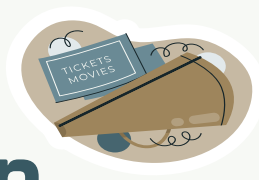
|       |             |
|-------|-------------|
| MSE   | 3.6737e+6   |
| MAE   | 1.06433e+08 |
| RMSLE | nan         |
| RMSE  | 1.91669e+08 |

## Deep Learning

|       |             |
|-------|-------------|
| MSE   | 3.47688e+16 |
| MAE   | 1.1276e+08  |
| RMSLE | nan         |
| RMSE  | 1.86464e+08 |



# Models After Data Transformation



## Stacked Ensemble

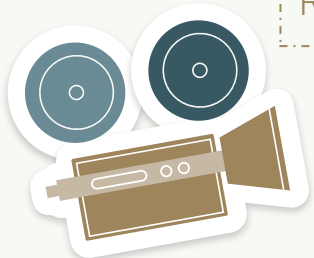
|       |        |
|-------|--------|
| MSE   | 0.1306 |
| MAE   | 0.2252 |
| RMSLE | 0.2006 |
| RMSE  | 0.3514 |

## Gradient Boosting Machine

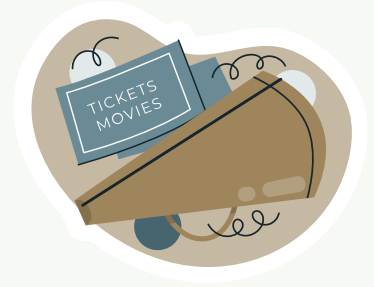
|       |          |
|-------|----------|
| MSE   | 0.1343   |
| MAE   | 0.2289   |
| RMSLE | 0.203783 |
| RMSE  | 0.3665   |

## Deep Learning

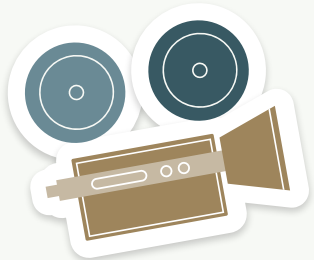
|       |          |
|-------|----------|
| MSE   | 0.143891 |
| MAE   | 0.228852 |
| RMSLE | 0.206746 |
| RMSE  | 0.379329 |



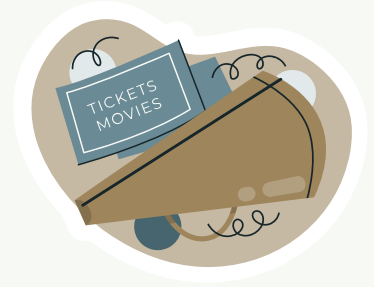
# Model Performance Insights



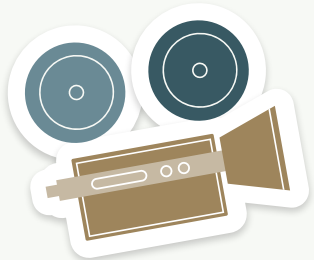
- ❖ Data cleaning (handling nulls, outliers) and transformations (normalization, encoding) played a critical role in increasing model accuracy
- ❖ For example, the **RMSE** for **Gradient Boosting Machine** dropped from approximately 1.916 to just 0.3665, indicating much better accuracy
- ❖ The **Stacked Ensemble** consistently outperformed the other models after transformations in each metric, making it the most accurate model for our task



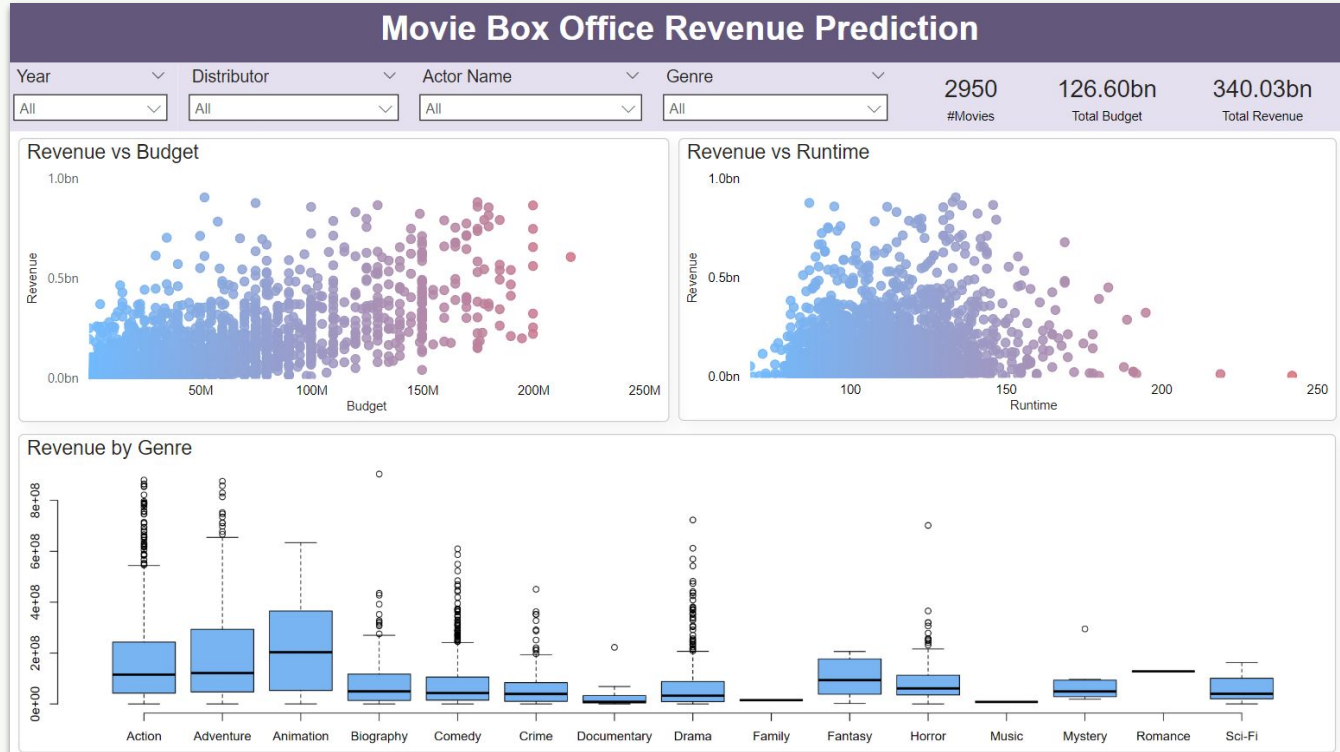
# Model Performance Insights



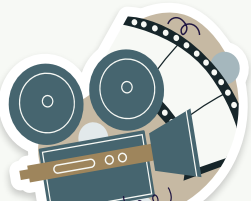
- ❖ **Stacked Ensemble** initially struggled with high errors, but transformations dropped the **RMSE** from 1.7782 to 0.3514, showing a significant reduction in prediction error
- ❖ **Deep Learning** performed comparably to **Gradient Boosting** post-transformation but may have been more sensitive to the dataset's size and noise.
- **RMSE** improved to 0.3793, matching **Gradient Boosting**, but the large initial **MSE** suggests a tendency to overfit during the early stages.



# Dashboard



Created with PowerBi and R

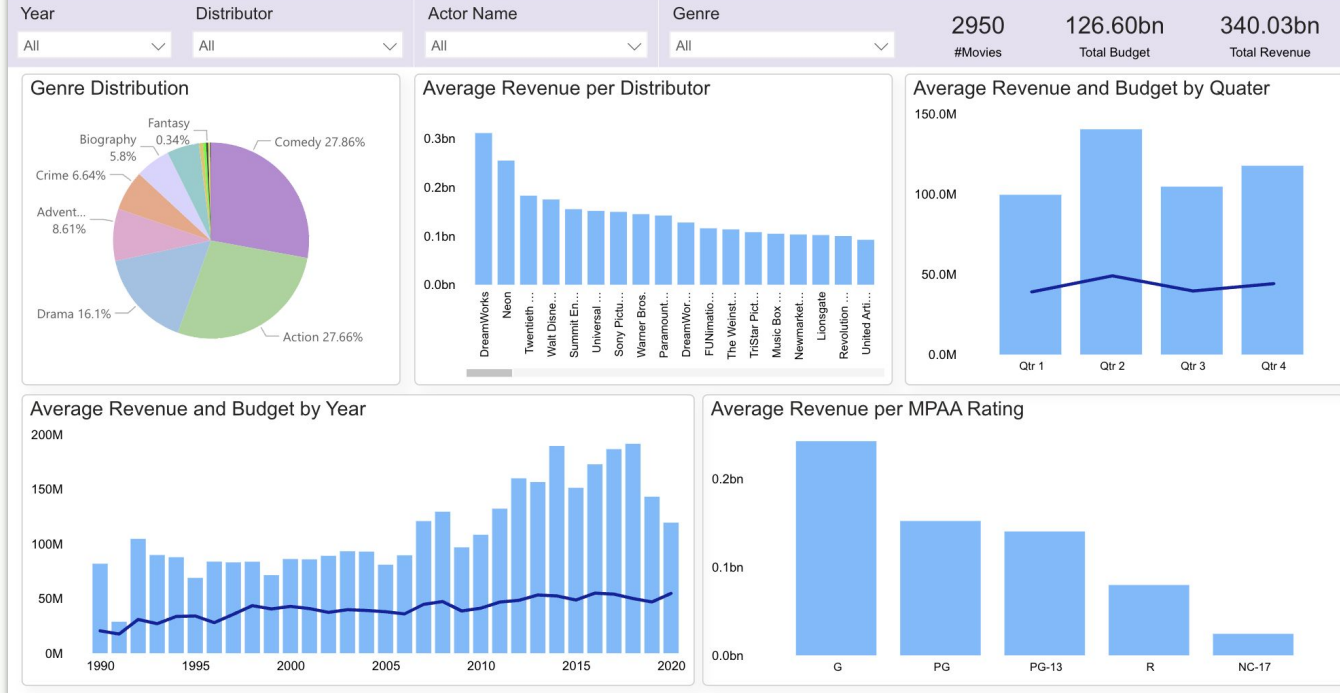




# Dashboard



## Movie Box Office Revenue Prediction





THANK YOU!

